

ARMAAN GUPTA

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🌐 armaan.world

Education

University of Waterloo

Bachelor of Computer Science + Minor in Artificial Intelligence (3.7+ GPA)

Sept. 2024 – Apr. 2029

Waterloo, ON

Technical Skills

Languages: Java, Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS, Swift, PHP, Ruby on Rails

Frameworks, Tools, & Libraries: React, Node.js, Git, AWS, TensorFlow, PyTorch, Firebase, Angular, Docker Next.js, npm, Flask, pandas, NumPy, Matplotlib, MongoDB, YOLO, OpenCV, Jupyter Notebooks, Google Colab, Tailwind, FastAPI, REST API, Vite, Three.js, Figma, Linux, Bash, GCP

Experience

WATonomous

July 2025 – Present

Software Engineer

Waterloo, ON

- Built an Automated Power Outage Notification System, leveraging **cloud-based services** and containerized microservices with Docker, reducing manual reporting delays by **80%**.
- Developed cloud-native APIs and automation scripts to integrate with SLURM and Kubernetes, enabling **parallel execution** of simulation jobs and increasing throughput by **300%**.
- Developed and containerized backend services with Docker, deploying to cloud and **HPC clusters** via Kubernetes

Environment Canada (ECCC)

May 2025 – Aug. 2025

Software Engineer

Toronto, ON

- Developed a climate reanalysis model pipeline by integrating **ML-based** interpolation techniques using **scikit-learn** and **PyTorch**, improving **spatial resolution** of missing environmental data by **35%**.
- Built an interactive web tool to explore **40+ years** of climate data across **50+** stations and **16 datasets** using Python, pandas, and D3.js, giving Canadians access to novel reanalysis datasets.
- Automated deployments and **version control** via **git**, **Docker**, and **CI/CD** internal practices across multiple projects.

LandingX

July 2023 – May 2024

CEO & Founder

Remote

- Engineered modern, high-performance web applications with React.js, Node.js, Three.js, and Tailwind CSS, delivering seamless **cross-device compatibility** for **1000s of users**.
- Deployed and optimized applications using Vercel and AWS, **reducing load times by 30%** through performance tuning and **CDN utilization**.
- Managed full-stack development workflows, leveraging Vite, npm, and Git for version control and deployment.

Projects

PrepPal | SQL, Python, JS, React, Tailwind, Flask, Google Cloud, OpenCV, OpenAI, Cohere, Recharts

🔗 [repo link](#)

- Developed a **full-stack** interview preparation assistant to provide interactive mock interviews and AI-driven feedback.
- Integrated OpenAI and Cohere APIs to formulate personalized GenAI feedback and **real-time response analysis**, improving user preparation by **40%**, resulting in **100+ unique users** within 2 weeks of launch.
- Designed and implemented a structured SQL database, efficiently storing user interview history, AI-generated feedback, and progress metrics.

ASD Prediction Model | Python, Colab, NumPy, Pandas, Matplotlib, sklearn, pickle

🔗 [repo link](#)

- Achieved **93%** cross-validation accuracy in detecting Autism Spectrum Disorder using **Decision Tree** and **Random Forest** models.
- Processed data with NumPy, pandas, Matplotlib, and SMOTE to handle imbalances and extract insights.
- Optimized models in Google Colab, leveraging scikit-learn for **hyperparameter tuning** and pickle for model storage.

Speed Estimation & Vehicle Tracker | Python, OpenCV, ultralytics, NumPy, Argsparse

🔗 [repo link](#)

- Built a **CNN-based** vehicle detection and tracking model capable of calculating vehicle speed, counting vehicles, and classifying types for **detailed traffic analysis**.
- Integrated real-time video processing with Supervision & OpenCV, leveraging **tqdm to optimize performance**.
- Streamlined model deployment and inference with requests, ensuring **efficient** and accurate processing for large-scale traffic datasets and obscure camera angles.